ITR: Learning and Measuring Perceptual Similarity

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Project Summary

Image retrieval has been an active research area for many years, but two fundamental problems remain largely unsolved:

- How best to learn users' subjective query concepts (search profiles), and
- How to measure perceptual similarity with significant accuracy.

The first problem concerns the completeness of formulating a search profile, e.g., how to formulate a query such as "animals" or "enemy vessels." The second problem concerns search accuracy, i.e., given a search profile, how to find all images that match that profile.

To tackle these two fundamental problems and to ensure that our solutions are scalable, this project's specific targets are as follows:

1. To develop novel *active learning algorithms* that learn a query profile quickly despite time and sample constraints. (E.g., Constructing an enemy-vessel profile quickly with meager data.)
2. To design *semi-automatic image annotation* and *annotation refinement* methods for assigning semantic labels to images in order to support *cross modality* query-concept learning and information retrieval.
3. To devise *perceptual distance functions* for improving visual search accuracy. For instance, once a search profile such as "enemy vessels" is learned, we would not want to miss any matching object in the surveillance database.
4. To conduct *validation* on developed learning algorithms with experimental data provided by colleagues at various institutions (including IBM research centers and Fine Arts Museums of San Francisco).

This project could have radical impact upon multimedia information retrieval. For example, just the work of *perceptual distance functions* can dramatically change how multimedia data are organized and indexed. One early evidence of such impact is that since May 2001, we have been invited to present preliminary results for the CIA, and at NSF sponsored workshops at INRIA and Harvard University; ACM and IEEE sponsored conferences; HP, IBM, and SONY research centers; Stanford University, UC Berkeley, UCLA, and other forums.